

Chapter 15

Further Macroeconomic Issues

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Recap

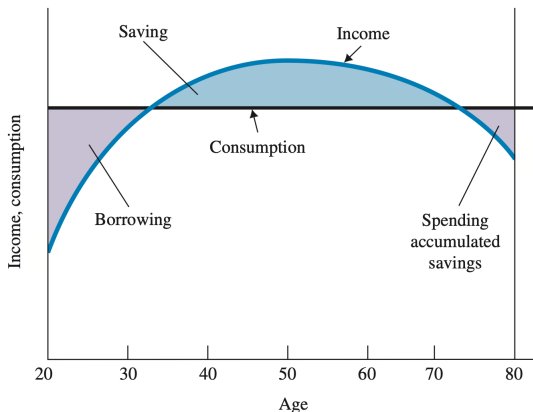
- In Chapters 8 through 14, we considered the interactions of households, firms, and the government in the goods, money, and labor markets.
- In these chapters, we assumed that household consumption C depends only on current income Y and that firms' planned investment I depends only on the interest rate r .
- In this chapter, we relax these assumptions and build a more realistic description of households' and firms' behaviors.

Household Behavior

- Household consumption depends on more than current income.
- Households determine consumption and labor supply simultaneously.
- The following factors affect household consumption and labor supply decisions:
 - Current and expected future real wage rates (life-cycle theory of consumption)
 - Initial value of wealth
 - Current and expected future nonlabor income
 - Interest rates
 - Current and expected future tax rates and transfer payments

The Life-Cycle Theory of Consumption

- A theory of household consumption that argues households make *lifetime* consumption decisions based on their expectations of *lifetime* income.



Permanent Income Hypothesis

- The key difference between the Keynesian theory of consumption and the life-cycle theory is that the life-cycle theory suggests that consumption and saving decisions are likely to be based not only on current income but also on expectations of future income.
- The average level of a person's expected future income stream is called **permanent income**.
- With this concept, we can summarize the life-cycle theory by saying that current consumption decisions are likely to be based on permanent income instead of current income.
- This means that policy changes such as tax-rate changes are likely to have more of an effect on household behavior if they are expected to be permanent instead of temporary.

The Labor Supply Decision

- Households play critical roles not only in consumption decisions, but also in labor supply decisions.
- What determines the quantity of labor supplied by a household?
 - Wage rate: substitution effect and income effect
 - Prices (real wage rate)
 - Wealth and nonlabor income

Interest Rate Effects on Consumption

- Recall that the interest rate affects a firm's investment decision.
 - This was a key link between the money market and the goods market, and it was the channel through which monetary policy had an impact on planned aggregate expenditure.
- Now we expand on this link: The interest rate also affects household behavior.
 - Interest rate is the reward (rate of return) to savings.
 - Change in interest rate: **substitution effect and income effect**

Government Effects on Consumption and Labor Supply

Taxes and Transfers

- The government influences household behavior mainly through income tax rates and transfer payments.
 - When the government raises (lowers) income tax rates, after-tax real wages decrease, lowering (raising) consumption.
 - A change in income tax rates also affects labor supply.
- Transfer payments are payments such as Social Security benefits, veterans' benefits, and general assistance to the needy.
 - An increase in transfer payments is an increase in nonlabor income, leading to a positive effect on consumption and a negative effect on labor supply.

A Possible Employment Constraint on Households

- Our discussion of the labor supply decision has so far proceeded as if households were free to choose how much to work each period.
 - But in reality, the amount of labor the household supplies is imposed on it from the outside by the workings of the economy (e.g. 8-hours per day).
 - Under these conditions, we do not expect changes in tax rates to influence labor supply behavior in the way we see in unconstrained markets.

NOTE: Keynesian Theory Revisited

- If there is unemployment and labor supply is constrained on the upside, Keynes is closer to being correct.
- In this case, income is directly determined by firms' hiring decisions and Keynesian theory of consumption becomes more persuasive.

Firm Behavior

- In discussing firm behavior previously, we assumed that planned investment depends only on the interest rate.
- However, there are several other determinants of planned investment.
- The following factors affect firms' investment and employment decisions:
 - Firms' expectations of future output (animal spirits)
 - Wage rate and cost of capital (interest rate)
 - Amount of excess labor and excess capital on hand
 - Inventory investment (change in the stock of inventories)

Excess Labor and Excess Capital Effects

- In practice, firms appear to hold what we will call excess labor and/or excess capital.
- Why might a firm want to employ more workers or have more capital on hand than it needs?
- To see why, suppose a firm suffers a sudden and large decrease in sales, but it expects the lower sales level to last only a few months.
 - The firm is likely to lower production in response to the sales change and decrease in production could mean that the firm could get rid of some workers and some machines.
 - Note that decreasing workforce and capital stock can be costly for a firm (**adjustment costs**).

Inventory Investment

- The change in the stock of inventories.
- The inventory investment decision is quite different from the plant-and-equipment investment decision.
- Recall the distinction between a firm's *sales* and its *output*.

$$\begin{aligned} & \text{Stock of inventories (end of period)} \\ &= \text{Stock of inventories (beginning of period)} \\ &+ \text{Production} - \text{Sales} \\ \rightarrow & \underbrace{\Delta \text{ Stock of inventories}}_{\text{inventory investment}} = \text{Production} - \text{Sales} \end{aligned}$$

The Optimal Inventory Policy

- At each point in time, a firm has some idea of how much it is going to sell in the current period and in future periods.
- Inventories are costly to a firm because they take up space and they tie up funds that could be earning interest.
- However, if a firm's stock of inventories gets too low, the firm may have difficulty meeting the demand for its product, especially if demand increases unexpectedly.
- Therefore, the point between too low and too high a stock of inventory is called the *optimal level of inventories*.
- The optimal level of inventories is realized at which the extra cost (in lost sales) from lowering inventories by a small amount is just equal to the extra gain (in interest revenue and decreased storage costs).